

S5: Activity Worksheet

Activity Title

Workload Model Bake-off

Duration

25 minutes total

- 5 min: briefing and context
- 12 min: group design and parameter choices
- 5 min: compare candidate workload models
- 3 min: debrief and answer review

Instructions

Given the EShop API behavior summary, each group proposes a workload model with:

- Action mix (browse/search/cart/checkout)
- Arrival pattern or VU stages
- Think-time assumptions
- Success metrics and SLO checks

Worksheet Template

Each audience group fills in their own row live during the 25-minute activity.

Group	Workload Mix	Peak Load Plan	Think-time Assumption	Risks
A				
B				

Answer Key (Reference Solution)

- **Suggested realistic mix:** Browse 40%, Search 35%, Add to Cart 20%, Checkout 5% per user session, based on the workload model the team validated for EShop (see 01_Report/Report.md Section 7.2). For the Black Friday framing of this activity, a reasonable variant shifts a few points from Browse toward Add to Cart and Checkout (shoppers arrive with more purchase intent during a sale event), while keeping Checkout a clear minority of actions, since even during a sale most sessions still browse or abandon before completing an order.
- **Why this mix is plausible:** It mirrors a typical e-commerce funnel where most traffic is read-heavy (browse and search) and only a small share of sessions convert to a purchase. Randomized think time of 1 to 3 seconds between actions (Section 7.3) approximates human pacing instead of back-to-back machine-speed requests, which keeps the load shape realistic rather than artificially bursty.

- **Common mistakes to avoid** (drawn from the EShop traps documented in the team's verified study of the EShop source):
 - Picking a raw VU number (for example, "500 users") without first defining an action mix. The core message of this activity is that the workload model, not the VU count, determines whether a result is trustworthy.
 - Forgetting the `/api` rate limiter (200 requests per 15 minutes per IP). A group that designs a search-heavy mix without accounting for it will see a wall of HTTP 429s that look like a real failure but are actually a measurement artifact, unless the `LOADTEST=1` bypass is deliberately used for the measurement run.
 - Reusing a single shared login/token across every virtual user for cart and checkout actions. EShop locks an account after 2 wrong-password attempts, and its cart state is kept in server memory per user, so a shared account can produce contention or lockouts that a per-user account pool would not.
 - Assuming an AI-generated load script already encodes a realistic business mix. AI can produce a syntactically correct script quickly, but it cannot judge whether the action ratios inside it reflect how EShop's real users actually behave; that judgment has to come from the group.
- **Minimum SLO checks:** p95 latency, error rate, and endpoint-level outliers (already present above, kept as originally written).